BUSINESS INTELLIGENCE (SECP3213)

ALTERYX LAB EXERCISE 2: DATA INVESTIGATION & TIME SERIES

CHAPTER 4 (PART 3)

SEM 2, 2025/2026

STUDENT GUIDELINE

Objective:
Use Alteryx Designer to apply data investigation and time series tools to understand data patterns and forecast future trends.

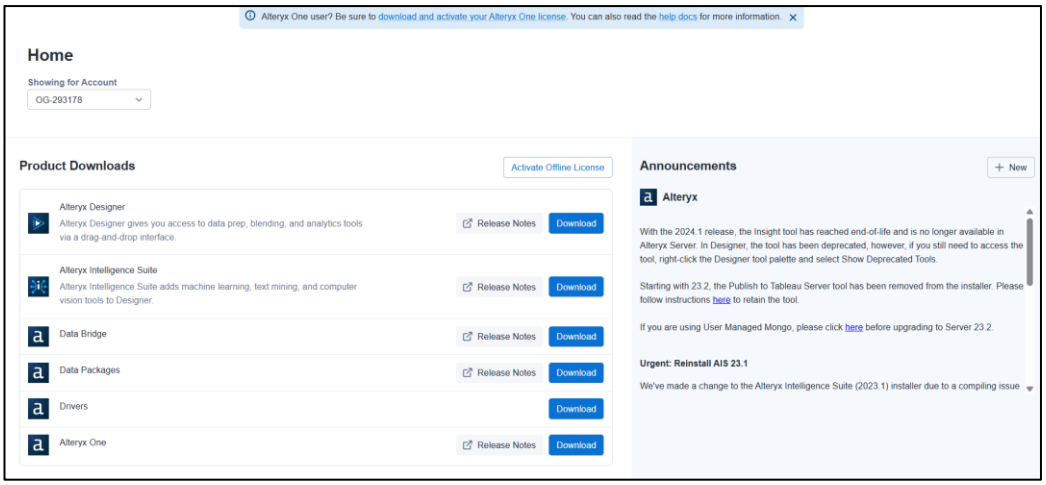
Datasets Used
a) SalesData.csv
b) CustomerDemographics.xlsx
c) ProductInfo.csv

Data Investigation Tools Installation

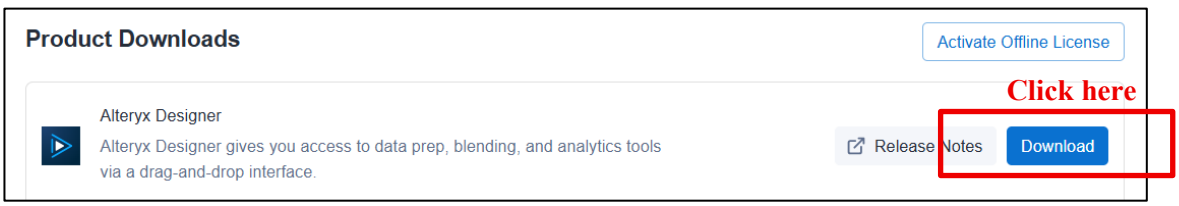
Before performing data investigation and time series analysis, the Alteryx Predictive Tools must be installed (**if the full toolset is not available, if toolset is available, you may skip these steps),** as they provide advanced analytics features such as:

- Association Analysis
- Contingency Table
- Scatterplot
- Time Series (ETS, ARIMA, TS Plot)

Step 1 : Access the following link : **https://us1.alteryxcloud.com/license-portal/**



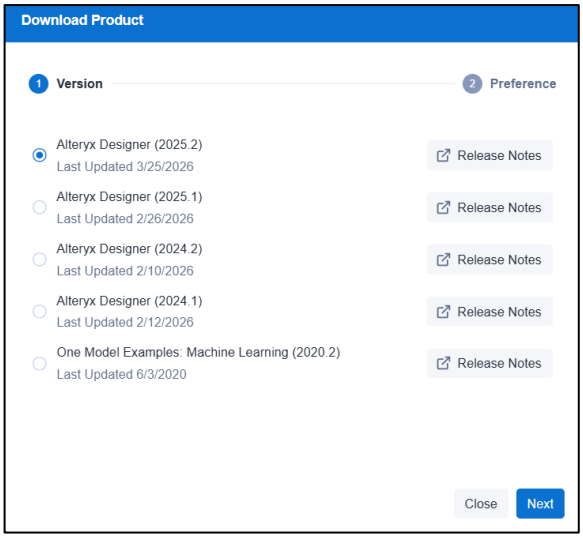
Step 2 : Access Alteryx Designer -> Download



Step 3: Choose the version that corresponds to your installed Alteryx Designer.

Step 4 : Then click Next and select the Alteryx Predictive Tools, ensuring you pick the correct option (Admin or Non-admin) based on your system setup.

Step 5 : Re-open your Alteryx Designer in your machine.



Scenario:
You are a data analyst at a retail company. Management wants to understand the customer demographics, purchasing trends, and forecast future sales to plan inventory.

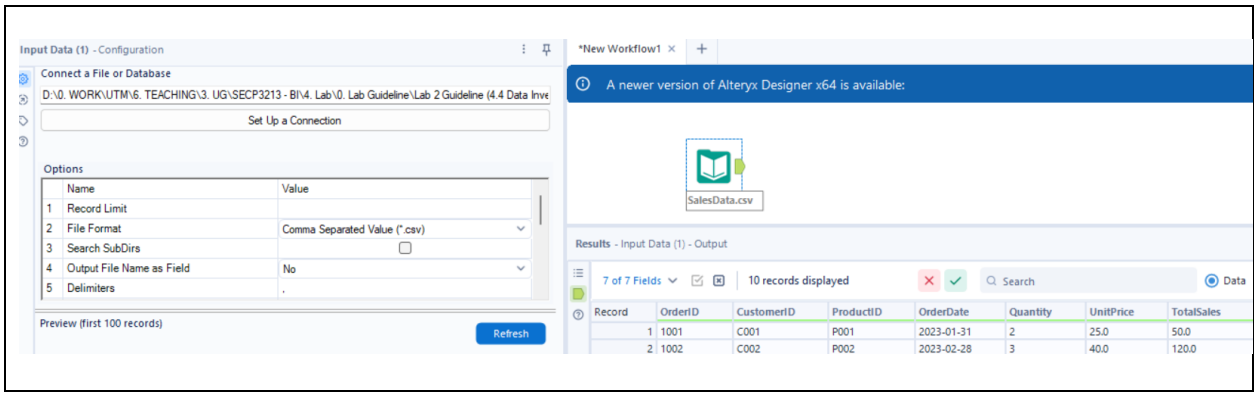
You are required to investigate the data quality, profile the customers, and analyze the sales pattern using Alteryx.

Data Investigation Tools Palette

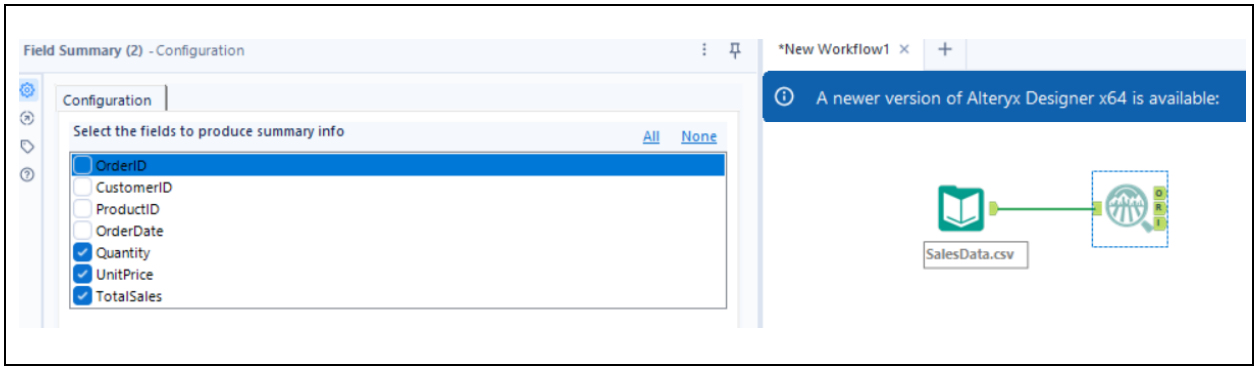
Exercise 1 Tools: Field Summary Tool
Purpose: Summarizes the structure and quality of each field.
Dataset: SalesData.csv

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a) Drag **Input Data tool** to load SalesData.csv.

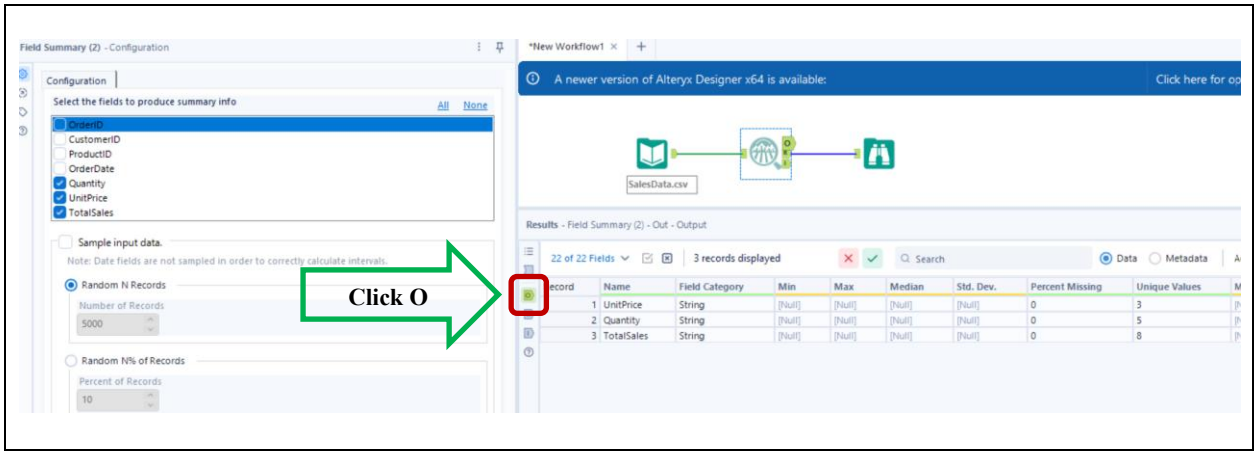


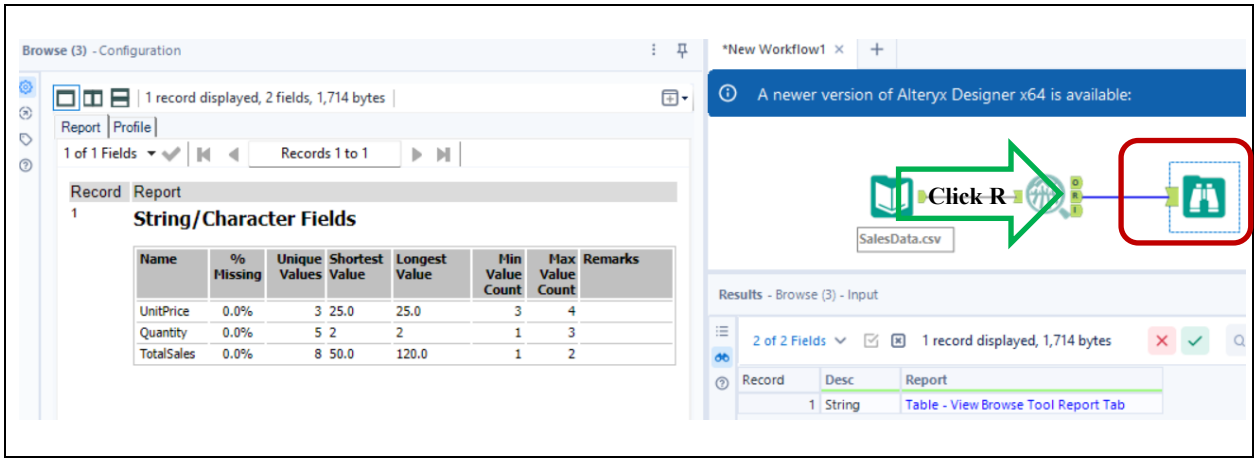
b) Drag the Field Summary tool and connect it to the Input Data.
   - Look for columns such as **Quantity, UnitPrice, TotalSales**.



c) Add a Browse tool to view the output.
   - Run the workflow.
   - **Observe**: Type, missing (Nulls), Min, Max, Mean, Standard Deviation, and Cardinality (Distinct Values).

d) Review the summary table that includes:
   - **O (Output – Final product):** Sends the processed dataset to the next tool and used for further analysis or workflow continuation.
     - **Example**: After profiling sales data:
       1. cleaned data
       2. transformed data
       3. filtered data
     - Continues to next step
   - **R (Report – Written Report):** Produces a summary report, often includes: statistics, distributions and data quality information.
     - **Example**:
       1. Average sales
       2. Missing values
       3. Min/max values
     - Useful for documentation and analysis reporting
   - **I (Interactive – Interactive dashboard/demo):** Generates interactive visual exploration and allows users to inspect data visually.
     - **Example**:
       1. charts
       2. interactive profiling
       3. drill-down inspection
     - Useful during exploratory data analysis (EDA)





e) **Summary**:

- **Quantity and TotalSales** are numeric fields with full completeness.
- **UnitPrice** may have mild outliers.
- **TotalSales** shows variability aligned with quantity.
- O and I flags are present for TotalSales indicating its potential influence. Quantity and TotalSales are numeric; check for null values, outliers in UnitPrice, and ensure data types are correctly recognized.

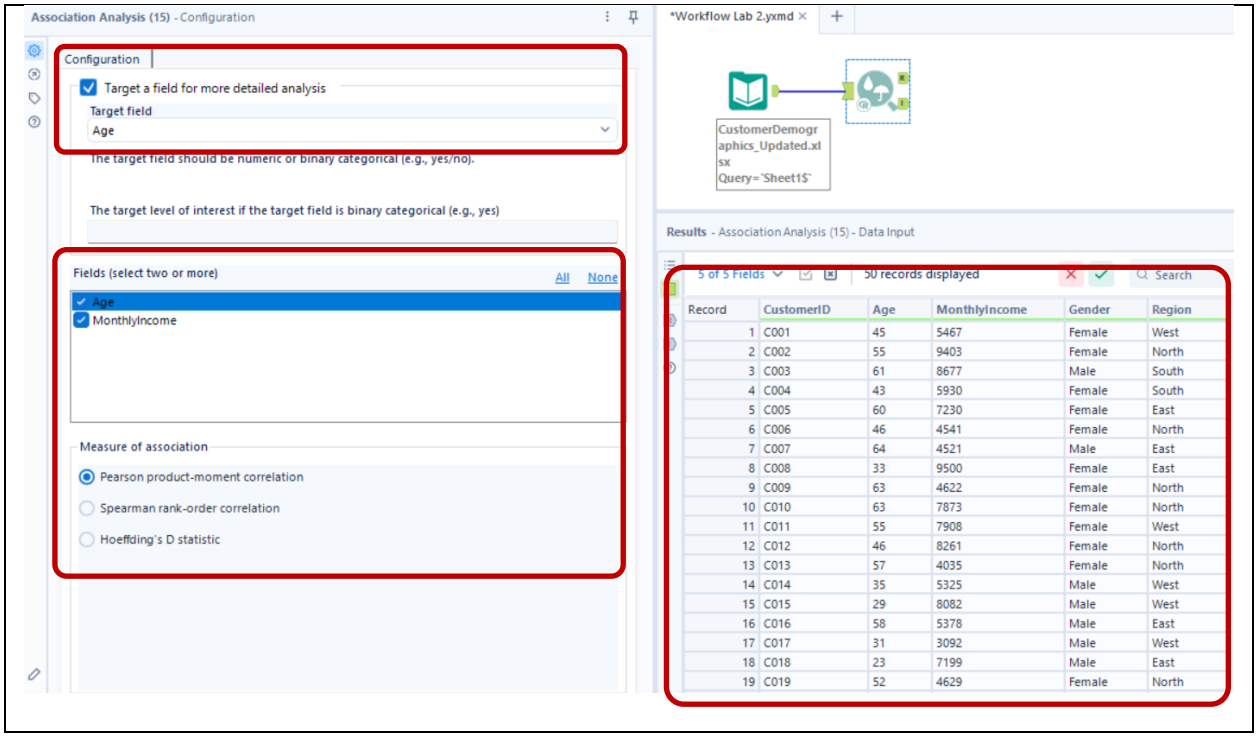
Exercise 2 Tools: Association Analysis Tool

Purpose: Finds relationships between categorical fields using support and confidence.
Dataset: CustomerDemographics_Upated.xlsx

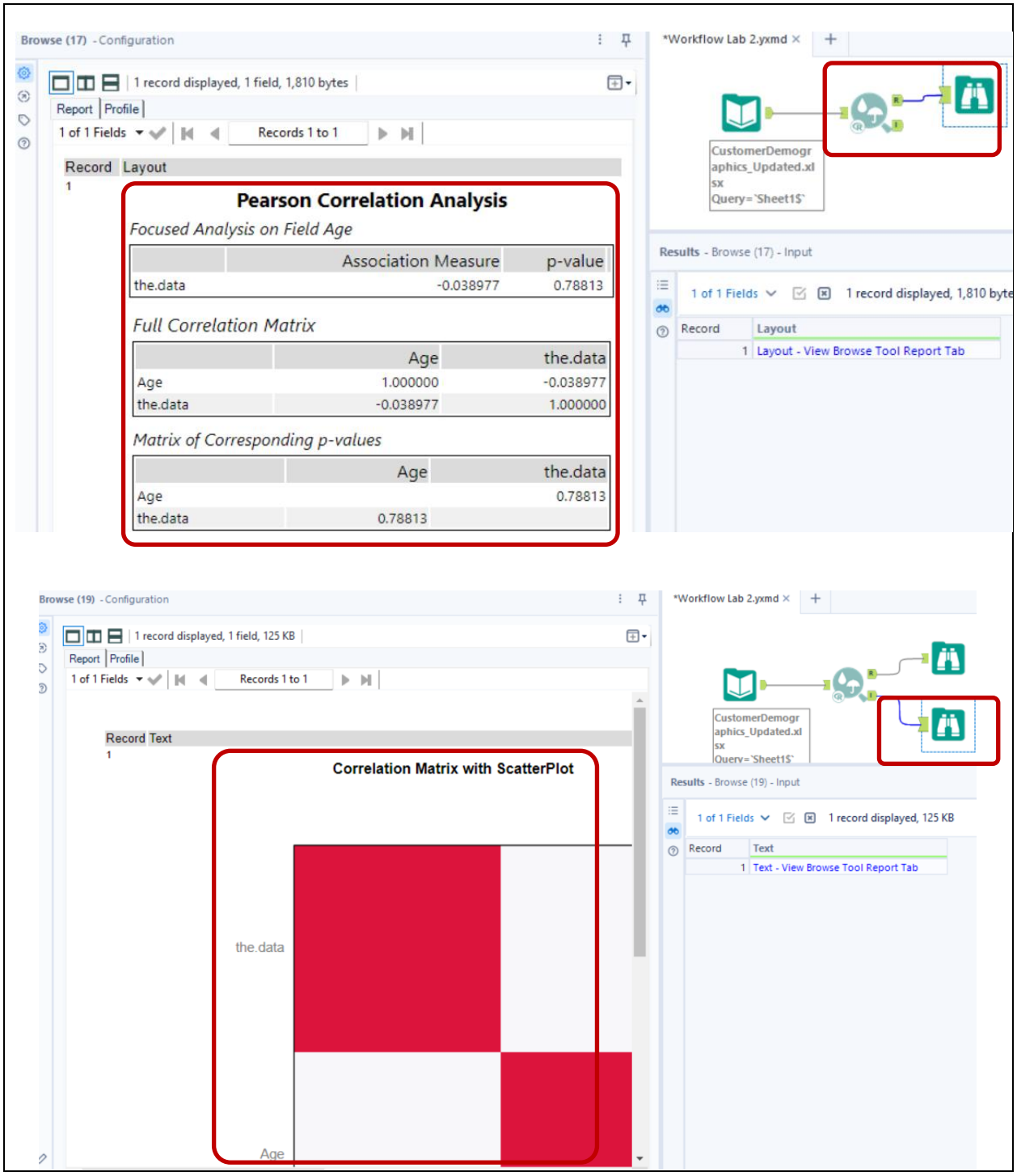
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a) Load CustomerDemographics_Updated.xlsx using Input Data tool.

- Drag the **Association Analysis tool** from the Data Investigation palette.
- In configuration, √ **Target** a field for more detailed analysis.
- Choose Age as the target field.
- In the list of available fields, √ **Age, MonthlyIncome**.
- Select **Pearson** as the association method.


Association Analysis (15) - Configuration
Configuration
Target a field for more detailed analysis
Target field
Age
The target field should be numeric or binary categorical (e.g., yes/no).
The target level of interest if the target field is binary categorical (e.g., yes)
Fields (select two or more)
All
None
Age
MonthlyIncome
Measure of association
Pearson product-moment correlation
Spearman rank-order correlation
Hoeffding's D statistic
*Workflow Lab 2.yxmd
CustomerDemographics_Updated.xlsx
Query=`Sheet1$`
Results - Association Analysis (15) - Data Input
5 of 5 Fields
50 records displayed
Search
Record
CustomerID
Age
MonthlyIncome
Gender
Region
1 C001 45 5467 Female West
2 C002 55 9403 Female North
3 C003 61 8677 Male South
4 C004 43 5930 Female South
5 C005 60 7230 Female East
6 C006 46 4541 Female North
7 C007 64 4521 Male East
8 C008 33 9500 Female East
9 C009 63 4622 Female North
10 C010 63 7873 Female North
11 C011 55 7908 Female West
12 C012 46 8261 Female North
13 C013 57 4035 Female North
14 C014 35 5325 Male West
15 C015 29 8082 Male West
16 C016 58 5378 Male East
17 C017 31 3092 Male West
18 C018 23 7199 Male East
19 C019 52 4629 Female North

- Add a **Browse tool** to inspect the correlation result.



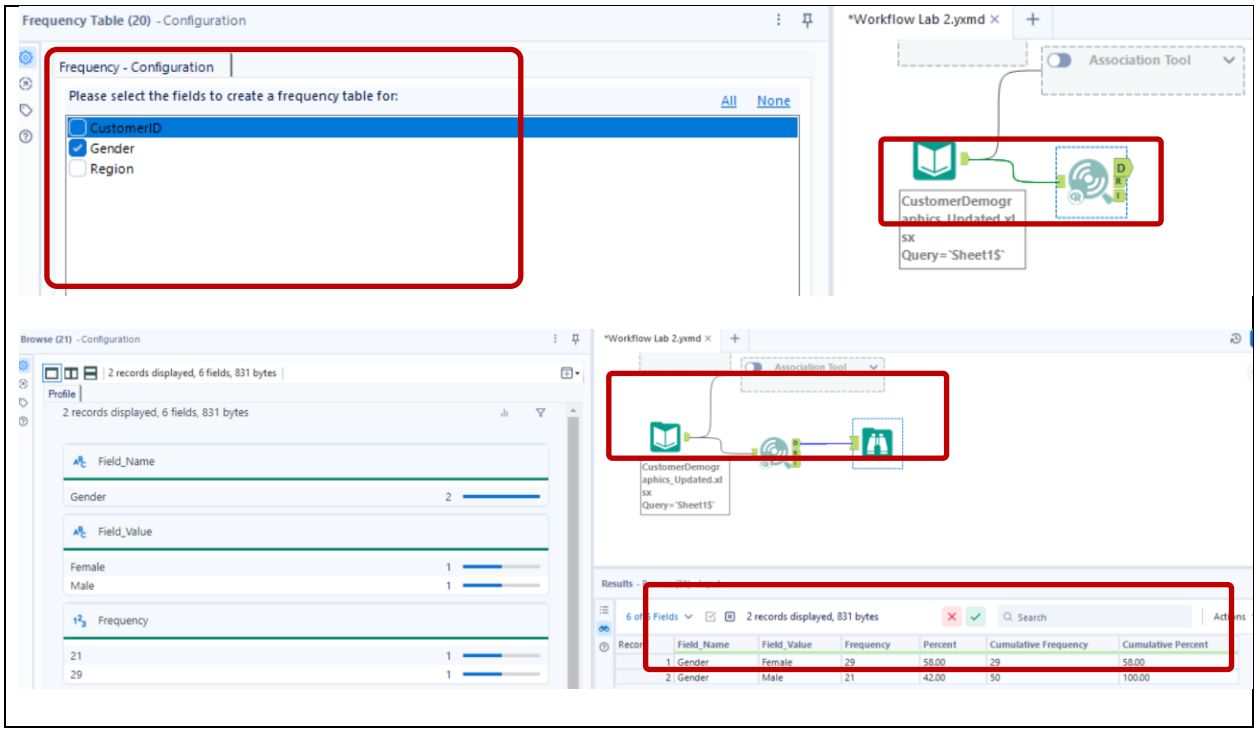
b) **Summary:**

- Displays correlation score between **Age** and **MonthlyIncome**.
- Positive value indicates income generally increases with age; negative suggests inverse relationship.

Exercise 3 Tools: Frequency Table Tool
Purpose: Displays frequency counts for unique values in a categorical field.
Dataset: CustomerDemographics_Updated.csv

a) Use **Input Data tool** to load CustomerDemographics_Updated.xlsx.

- Drag **Frequency Table tool.**
- In configuration, choose "**Gender**" as the grouping field.
- Add Browse tool to view results.



b) **Summary:**

- Shows how many records fall under each **Gender** category.

Exercise 4 Tools: Contingency Table Tool

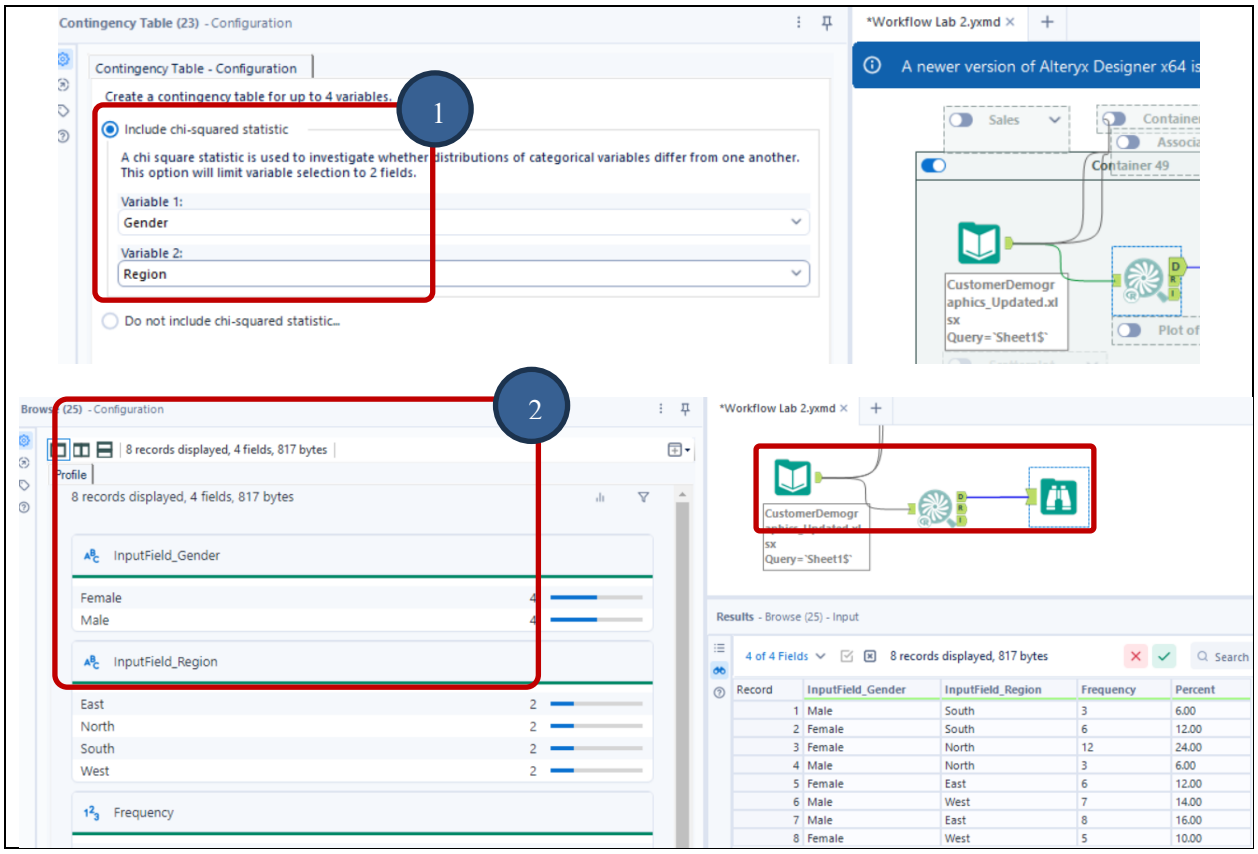
Purpose: Shows joint frequency distributions between two categorical fields.

Dataset: CustomerDemographics_Updated.csv

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a) Use **Input Data tool** to load CustomerDemographics_Updated.xlsx.

- Load the dataset and ensure **Gender** and **Region** fields are present.
- Add Contingency Table tool.
- Set **Gender** as the row and **Region** as the column variable.
- Add a Browse tool and run.



c) **Summary:**

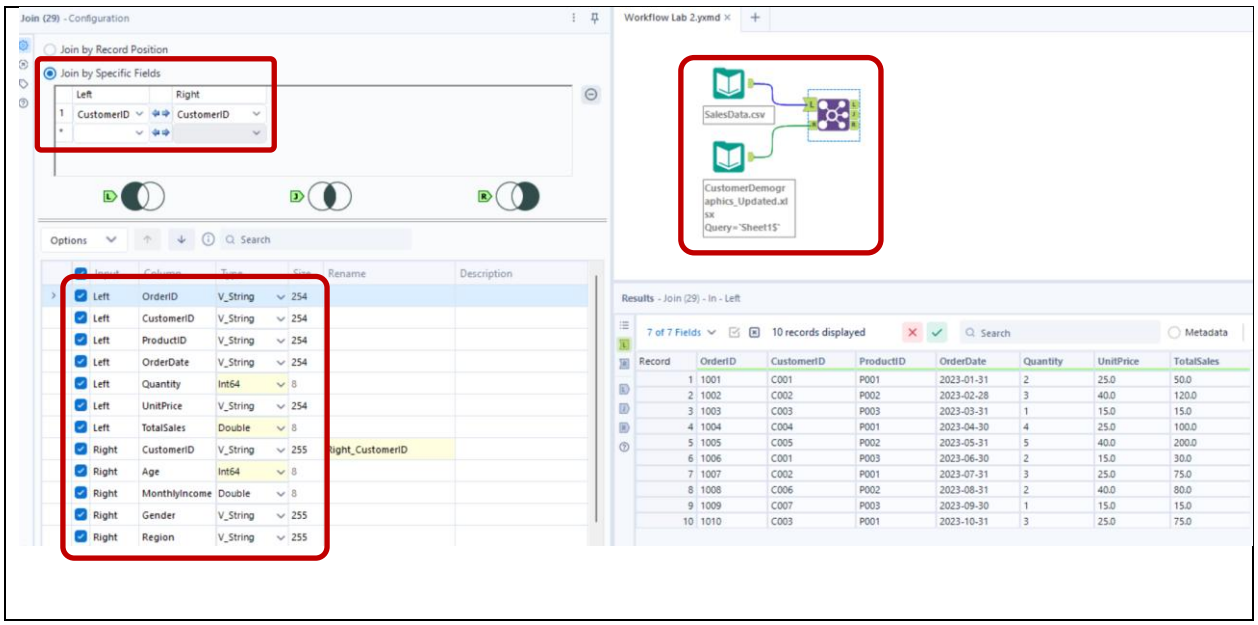
- Displays matrix showing how Gender is distributed across region.

Exercise 5 Tools: Plot of Means Tool

Purpose: Visualizes average values of a numeric variable across categories.
Dataset: SalesData.csv (joined with CustomerDemographics.xlsx)

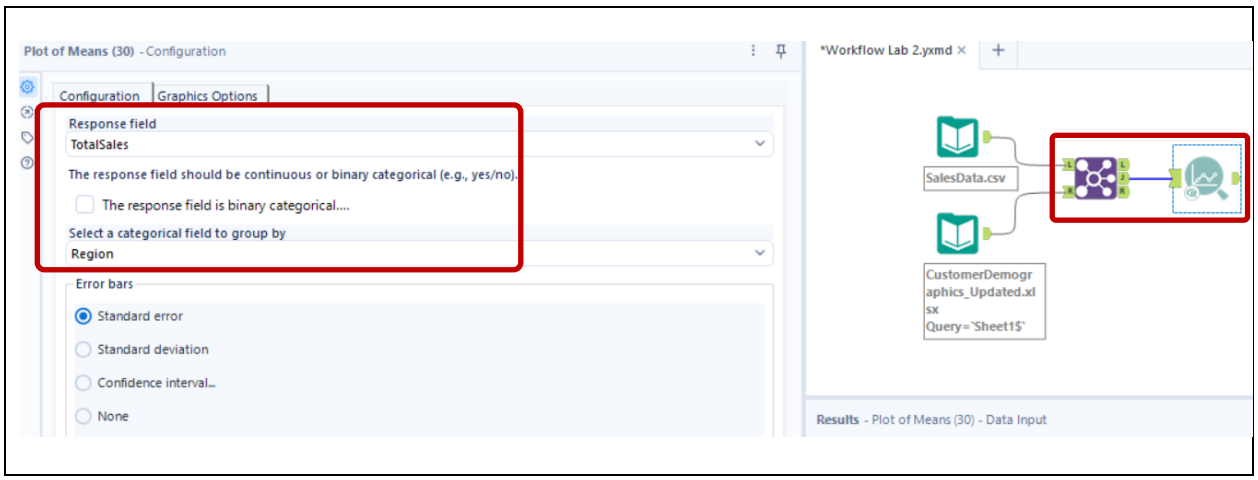
a) Use **Join** tool to combine SalesData with CustomerDemographics_Updated using CustomerID. Set data type:

- **Quantity** → Int64
- **TotalSales** → Double
- **Age** Set → Int64

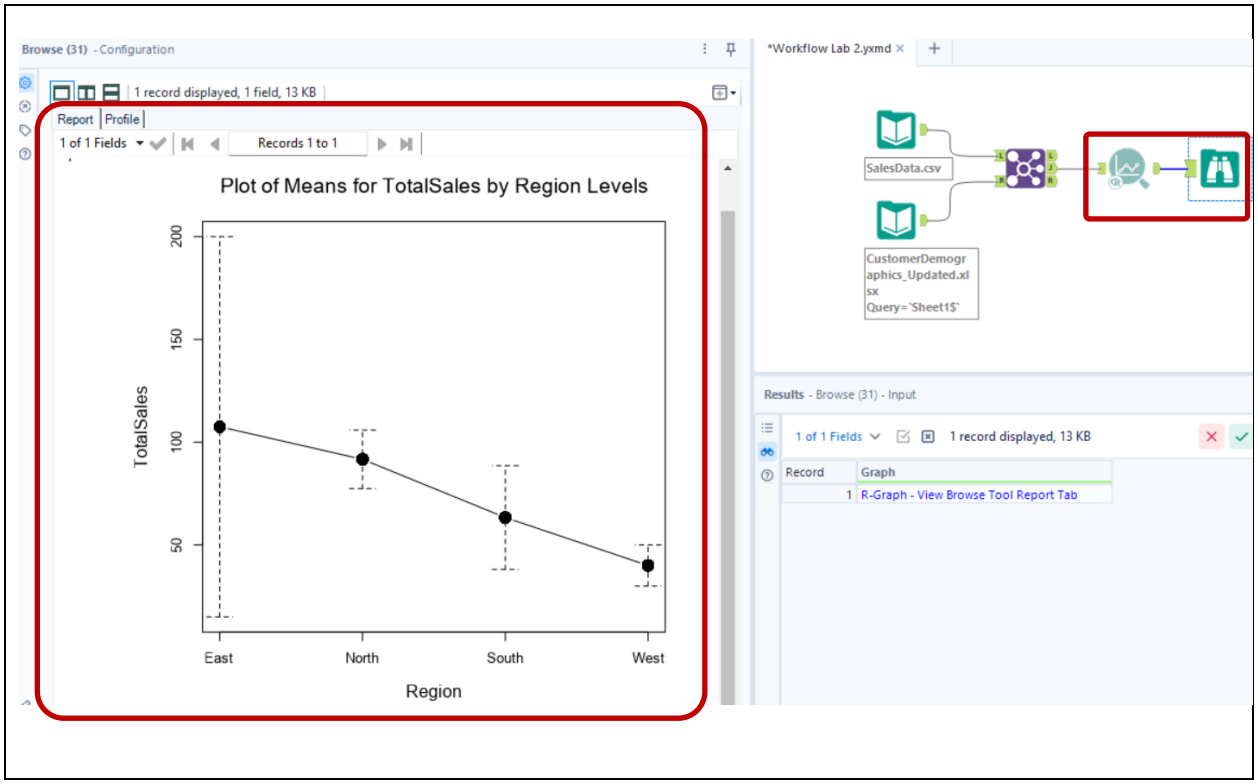


b) Add Plot of Means tool.

- Group by Region, Measure by TotalSales.
- Add Browse tool to visualize.



c) Add **Browse** tool to visualize.



d) **Summary:**

- Chart showing average TotalSales by Region.

Exercise 6 Tools: Scatterplot Tool

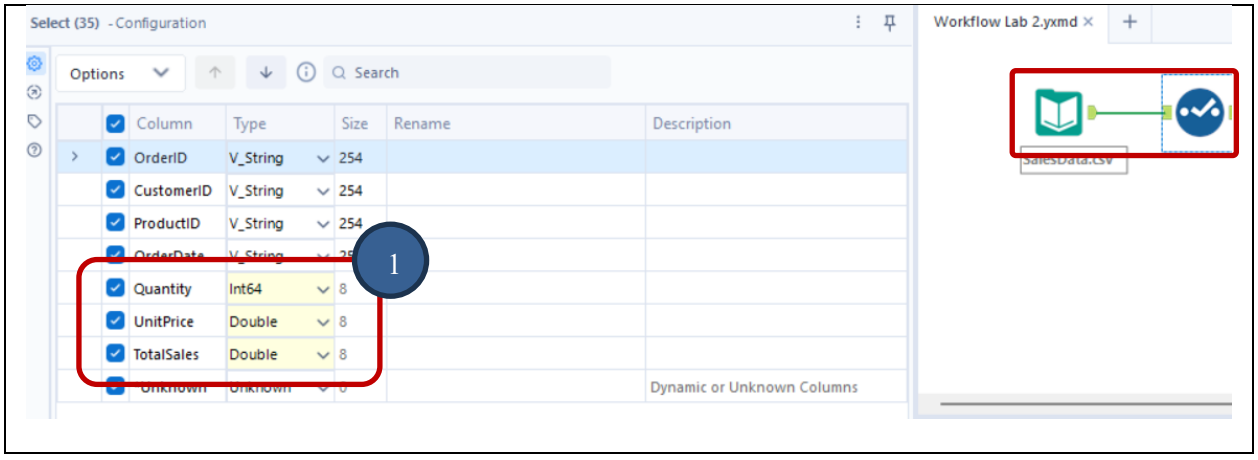
Purpose: Visualizes the correlation between two numeric variables.

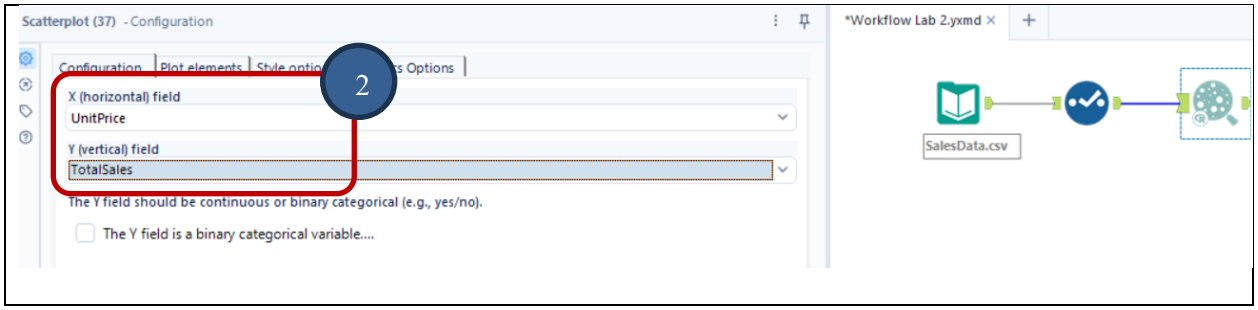
Dataset: SalesData.csv

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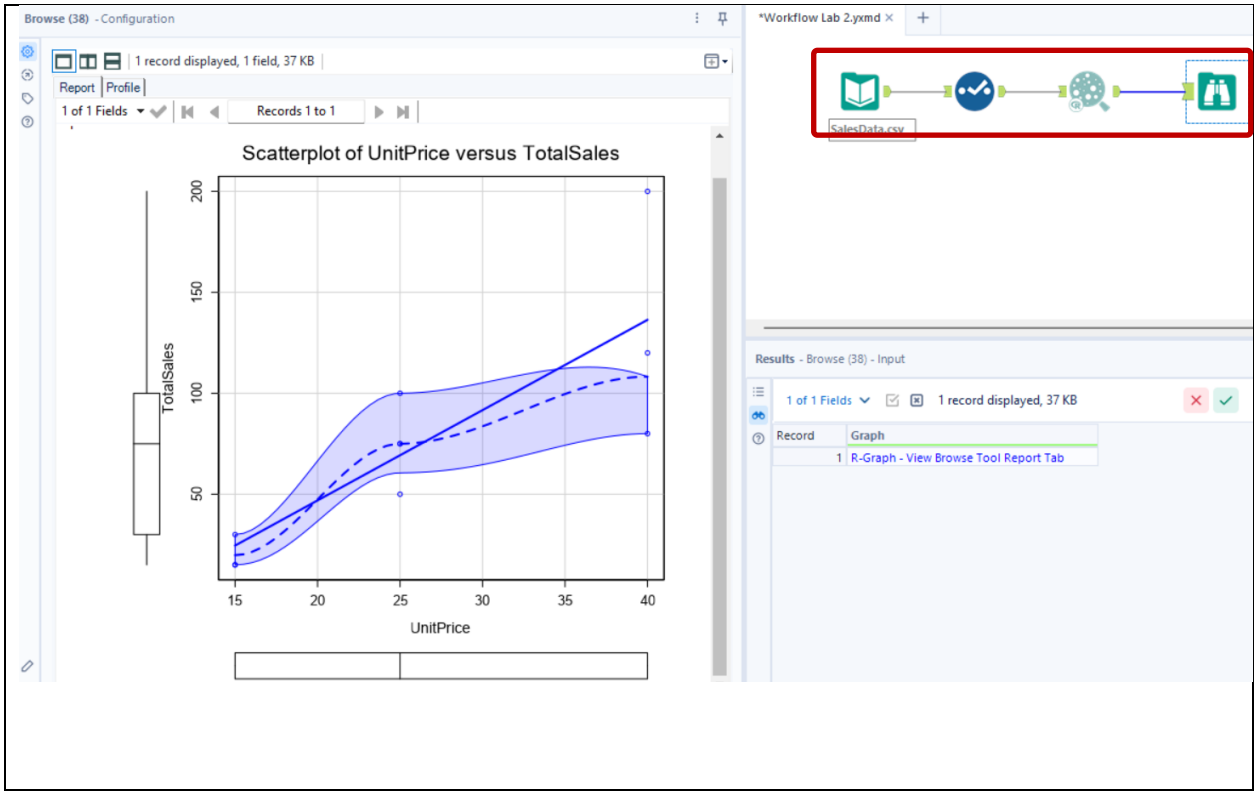
a) Load SalesData.csv and prepare the data for analysis.

- Using **Select Tool**, change the data type below:
  - **Quantity** → Int64
  - **UnitPrice, TotalSales** → Double
- Add Scatterplot tool.
  - Configure: X-Axis = UnitPrice, Y-Axis = TotalSales.





b) Add **Browse Tool** to display the chart.



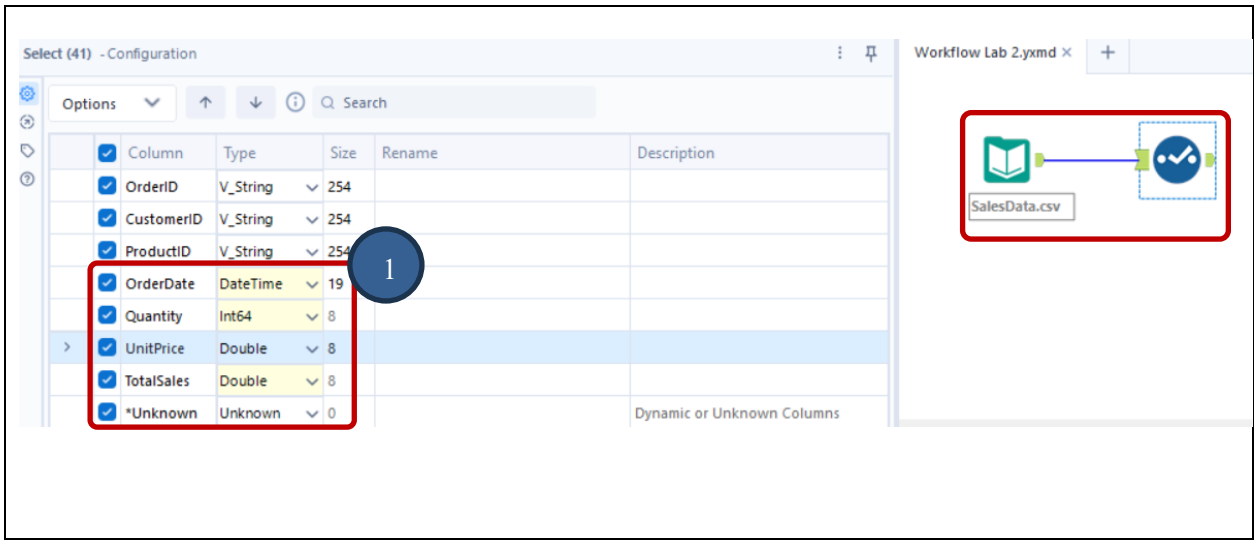
e) **Summary:**

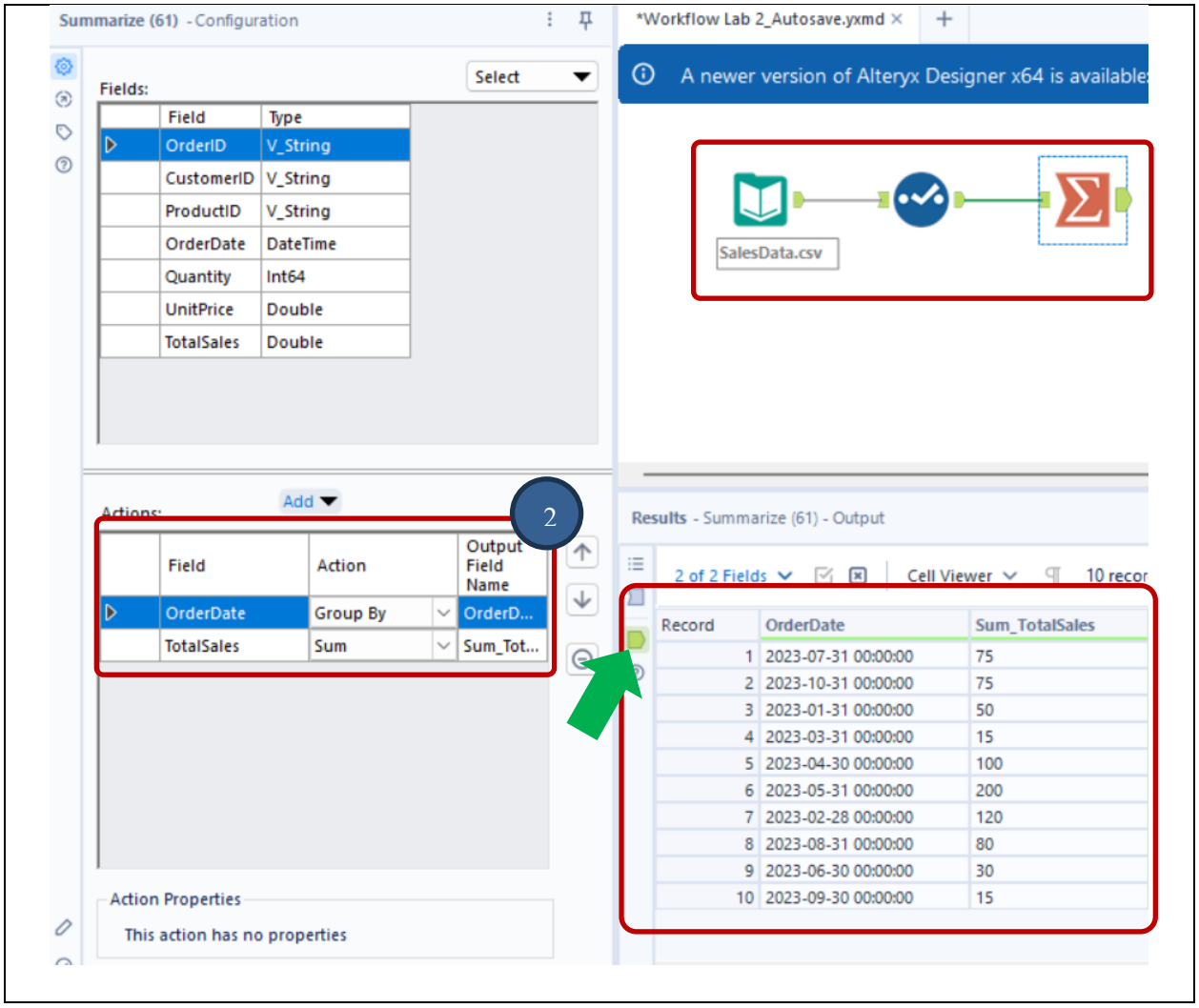
- Scatterplot that shows linear correlation between price and sales.

Time Series Tools Palette

Exercise 7 Tools: Time Series Plot Tool
Purpose: Visualizes trends in time series data to help identify patterns, seasonality, or anomalies in sales.
Dataset: SalesData.csv

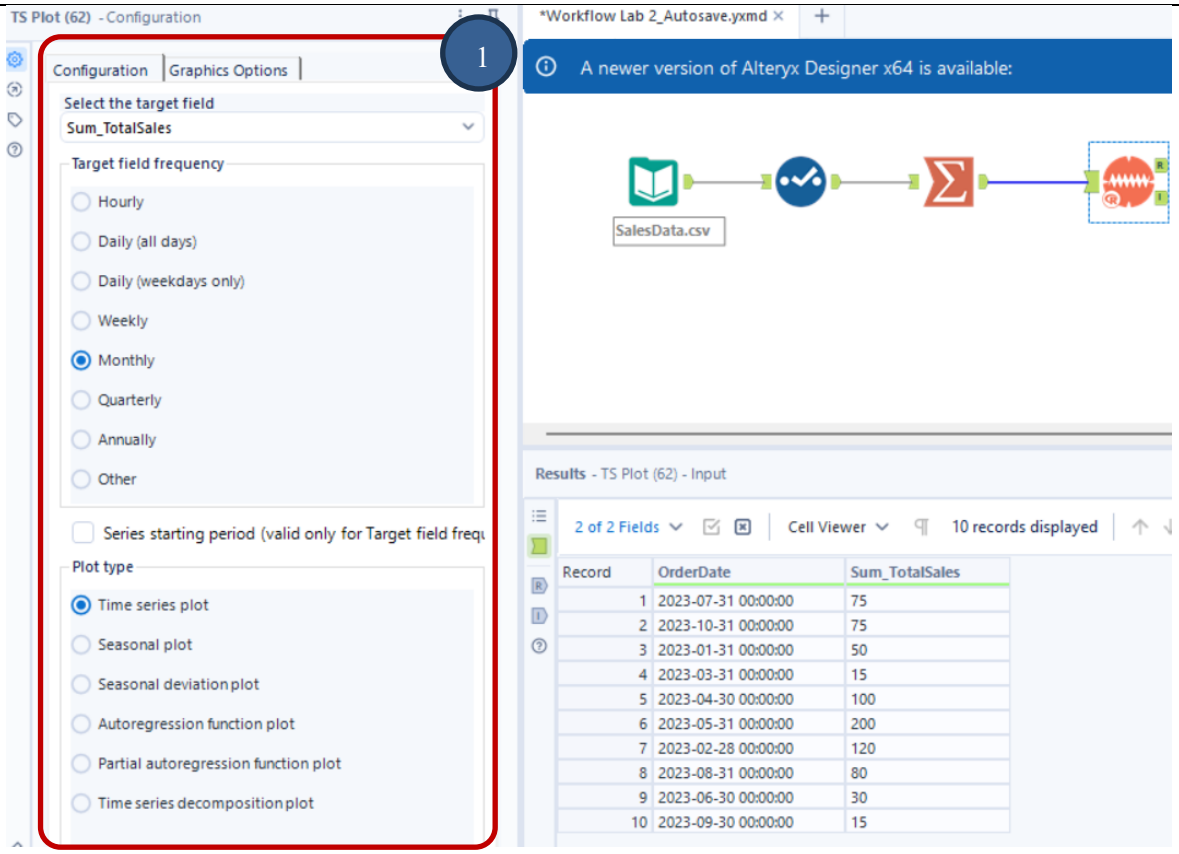
a) Use Input Data to load SalesData.csv.
- Using select tool, make sure **OrderDate** is in datetime format *(use DateTime tool if needed)* **UnitPrice** and **TotalSales** in double data type.
- Use a **Summarize Tool** to group **TotalSales** by month using **OrderDate** *(This will produce total sales per month)*
  - Group By → OrderDate
  - Sum → TotalSales



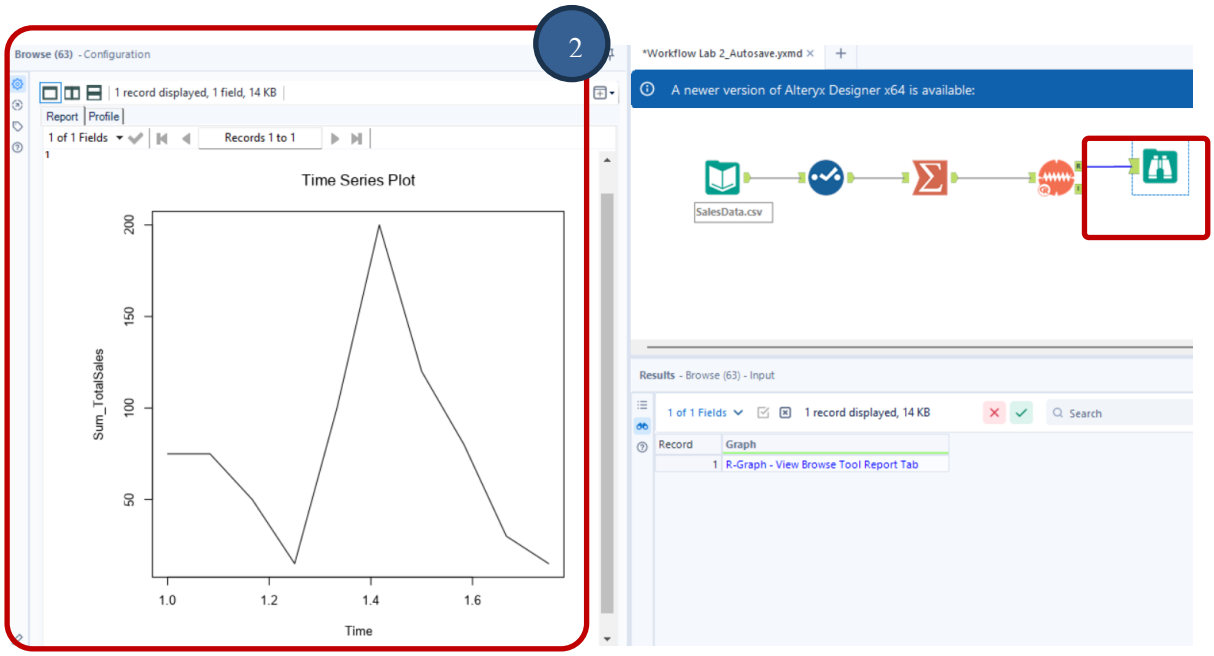


b) Add the **TS Plot tool** and connect it to the summarized data.
- Set **Sum_TotalSales** as the Target Field
- Choose **Monthly** for Target Field Frequency
- Select **Time series plot** as the Plot Type.
- Add Browse tool and run.

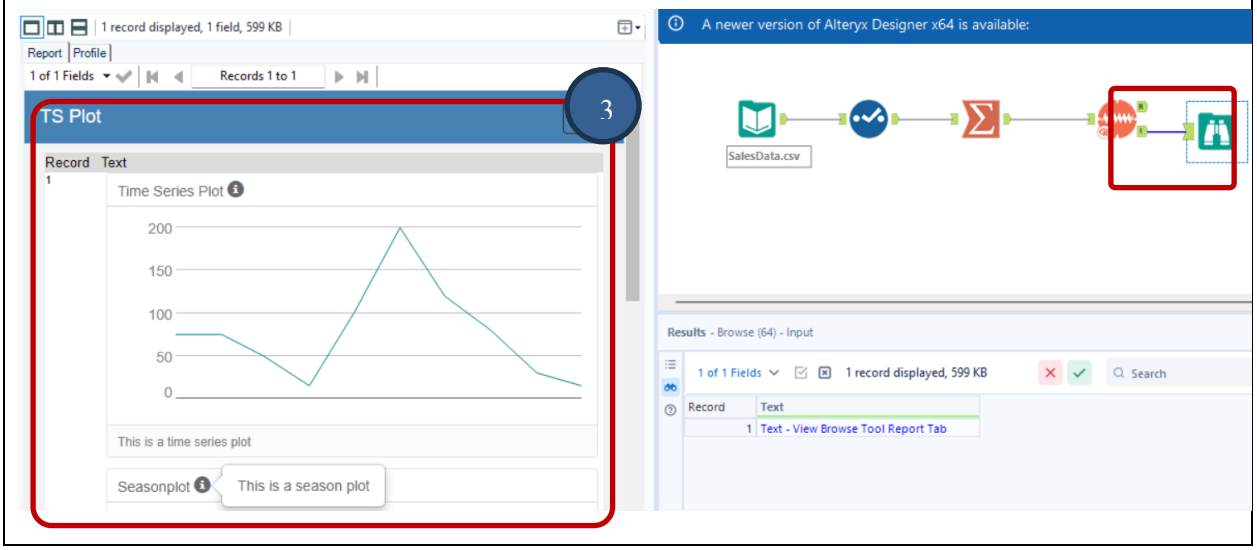
c) The TS Plot tool outputs have two anchors:
- R (Report): This produces a visual plot, which can be rendered in reports.
- I (Interactive): This provides the data table used in the plot for further analysis. These outputs help users to distinguish between report-ready visuals and raw trend data.



-Using R anchor-



-Using I anchor-



d) **Summary:**

- Displays a monthly trend of TotalSales over time.
- We should interpret if sales are increasing, seasonal, or declining over time.
- Displays an upward/downward trend in TotalSales over time.

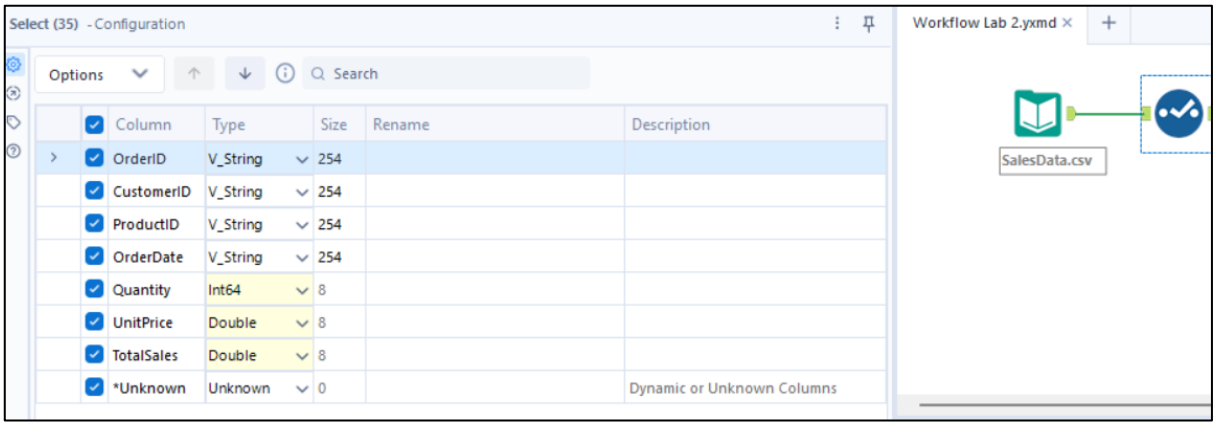
Exercise 8 Tools: ETS Tool

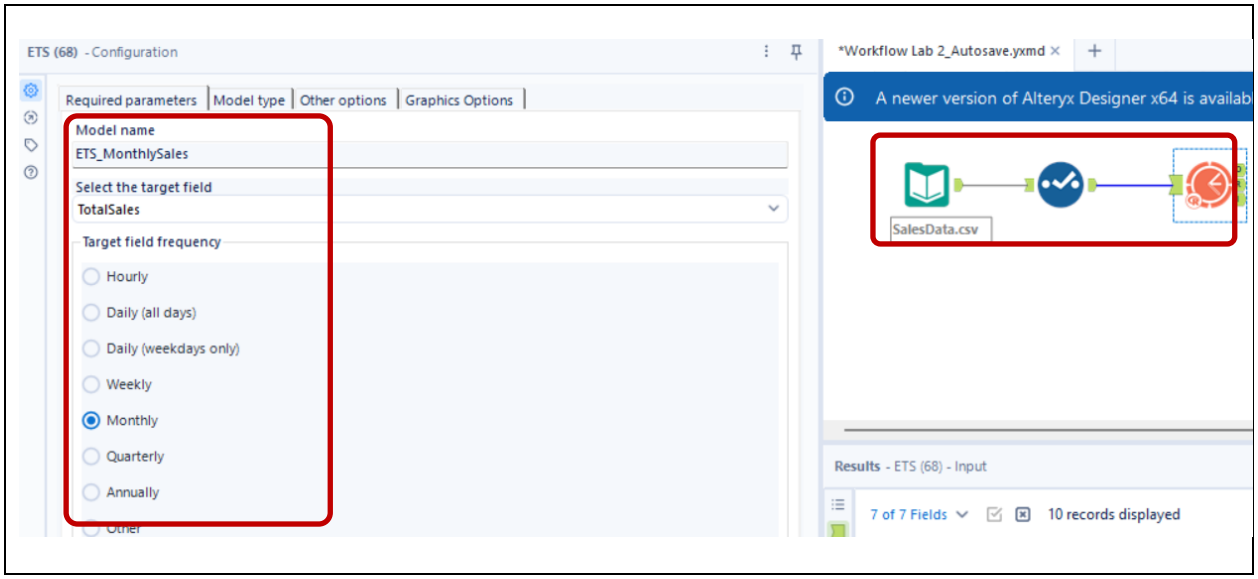
Purpose: Applies exponential smoothing to forecast future values based on past trends, accounting for seasonality.
Dataset: SalesData.csv

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a) Use the same input and DateTime-prepared data as in previous step.

- Add ETS Tool
- In configuration:
  - Set Model Name: ETS_MonthlySales
  - Target Field: TotalSales
  - Frequency: Monthly



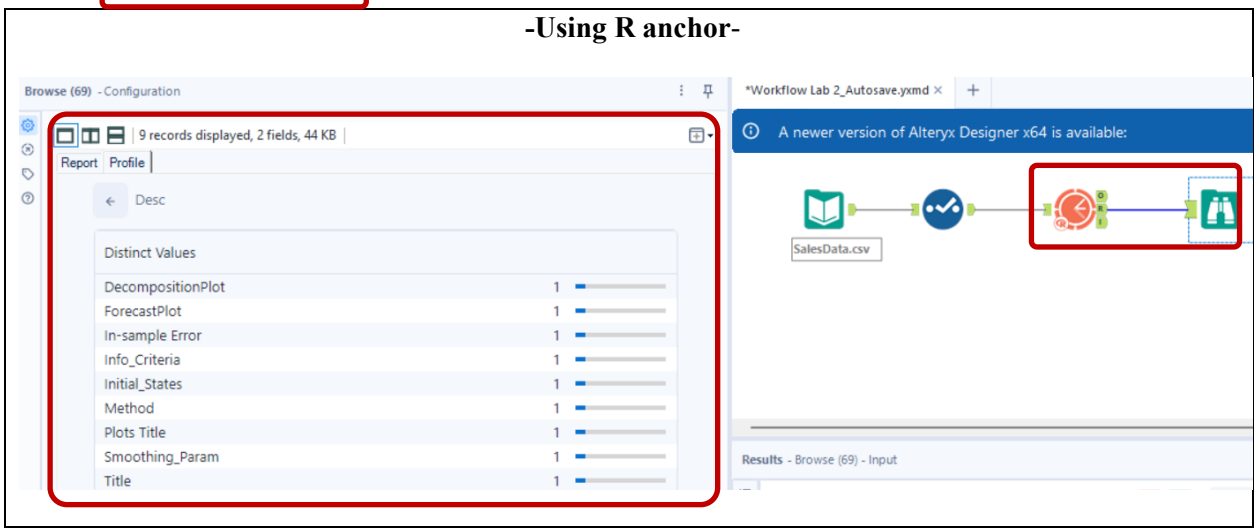


b) Add Browse tool and run.

c) The ETS tool produces multiple outputs:

- **R (Report):** A chart visualizing forecasted values, useful for dashboard and reporting (*it is a metadata report and it's what is passed into the Render Tool if we are building dashboards or documents).*
- **I (Interactive):** Tabular view of the forecast points with prediction intervals.
- These outputs help students interpret sales projections and understand the reliability range.

-Using R anchor-



d) Use the I (Interactive) output instead:

- Connect the green output anchor labeled I from the ETS Tool into the Browse Tool.
- This will show the actual forecast table — dates, predicted sales, and intervals.

e) Use the R (Report) output only if:

- The plan to include a chart in a formatted report using the Render Tool.

f) **Summary:**

- Forecast line with 95% confidence interval showing predicted future sales.
- Use **R** to display the graph and **I** to examine the data points behind the forecast. Forecast line with 95% confidence interval showing predicted future sales.

Exercise 9 Tools: ARIMA Tool

Purpose: Uses statistical modeling to capture trends, seasonality, and noise in a time series, producing forecasts.

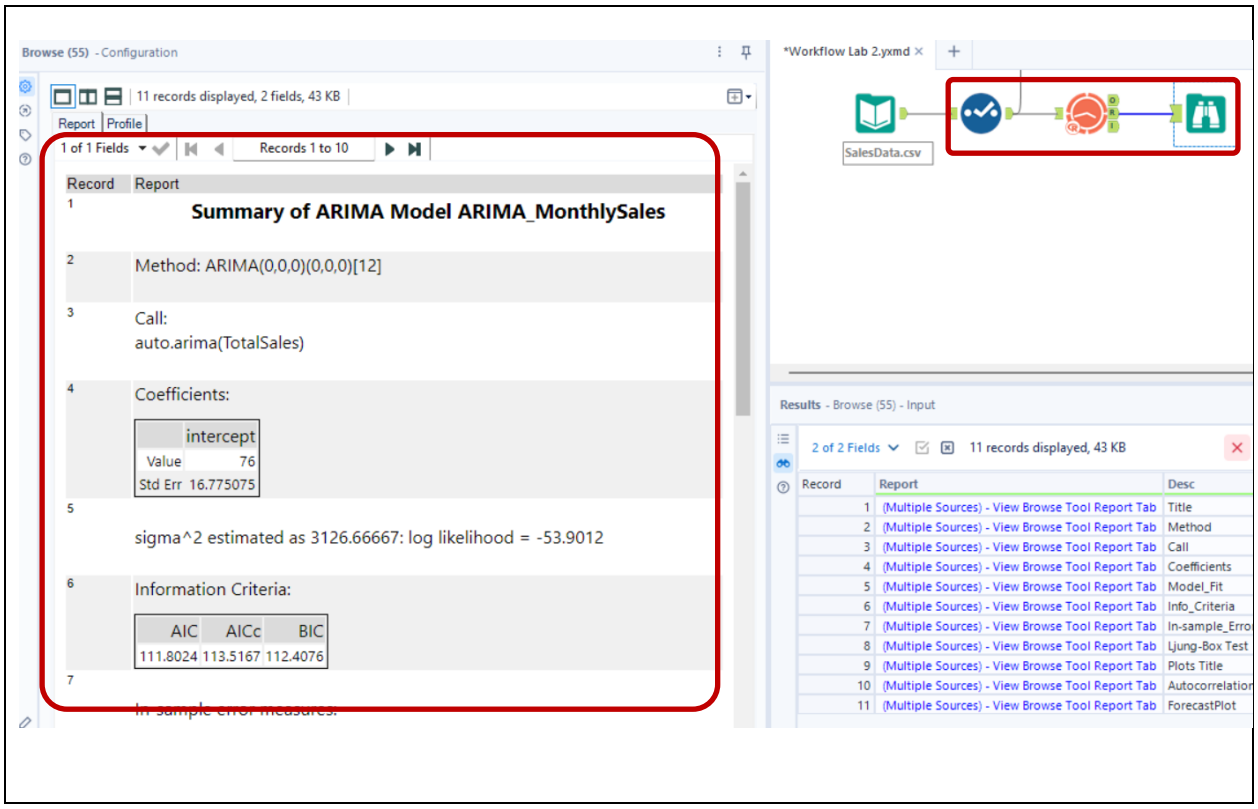
Dataset: SalesData.csv

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a) Use se same input and config in previous step.

- Add ARIMA tool. Set configuration:
  - Set Model Name: ARIMA_MonthlySales
  - Target Field: TotalSales
  - Frequency: Monthly.

b) Add Browse and run.



c) **Summary:**

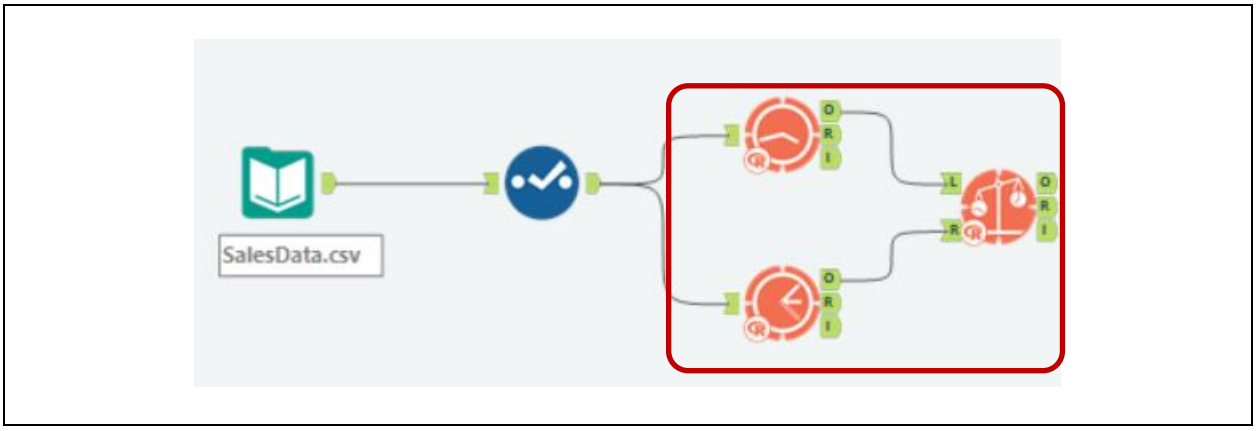
- ARIMA (Auto-Regressive Integrated Moving Average) produces two outputs:
  - R (Report): Graphical report showing the forecast trend.
  - I (Interactive): A table containing predicted values and confidence intervals.
- These outputs help us to see how ARIMA models past patterns and use them to project future values.
- Forecasts showing smoothed trend lines that adapt to previous seasonal behavior. Use R for the visual chart and I to extract or analyze the forecasted values.

Exercise 10 Tools: TS Compare Tool

Purpose: Compares the accuracy of multiple forecasting models (ETS vs ARIMA) to help choose the most reliable one.

Dataset: Outputs from ETS and ARIMA tools

a) Ensure both ETS and ARIMA tools have successfully generated model outputs.
- Set valid model names in both (e.g., ETS_MonthlySales, ARIMA_MonthlySales).
- Connect the model output anchors from ETS and ARIMA into TS Compare.
- Add Browse tool to inspect comparison.



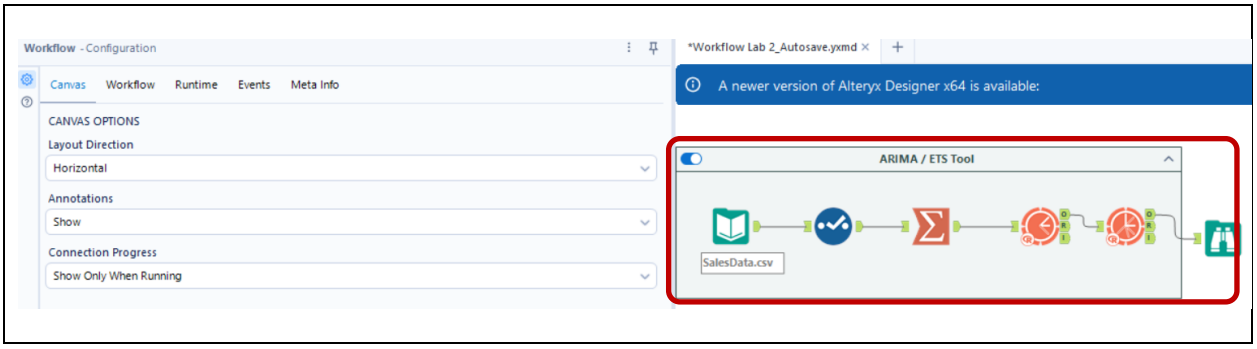
b) **Summary**
- Helps to evaluate which forecasting model fits the data better using metrics like MAPE and RMSE. Clarifies model performance based on accuracy.

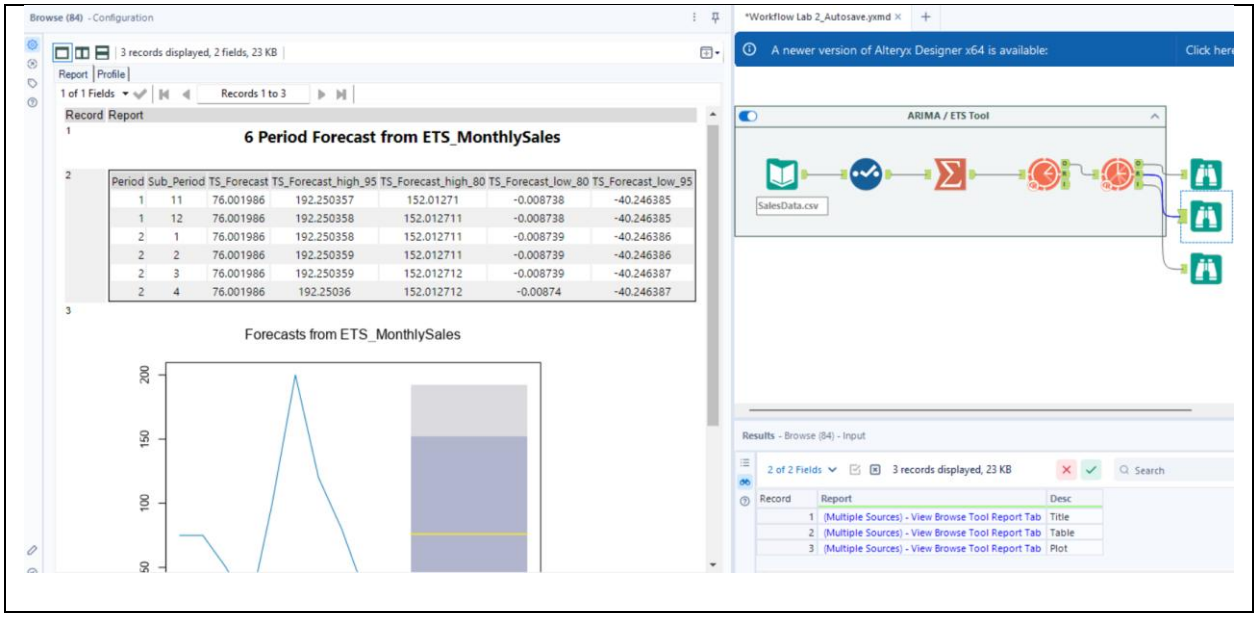
Tools: TS Forecast Tool

Purpose: Generates a forecast table from the selected time series model (ETS or ARIMA) for planning.

Dataset: Output from the best-performing model

a) Choose the best model output (from ETS or ARIMA).
b) Connect it to the TS Forecast tool.
c) Add Browse tool to see final forecast.





d) **Summary:**

- The tool extracts a detailed forecast table, including the predicted values and confidence intervals. The output is tabular (not graphical), listing: Forecasted periods (dates), Predicted values and Lower and Upper Confidence Intervals (This is ideal for making decisions about inventory, budgeting, or staffing).
- Forecast table with future predicted TotalSales values for each month, along with confidence ranges.

Final Reflection:

The series of exercises in this lab are designed to address the full business scenario:

Exercises 1–6 assess and explore the data quality and customer patterns

Exercises 7–9 generate and compare time series forecasts

Exercises 10–11 help validate the best model and output actionable forecast tables

Together, these above execises provide insights for retail managers to better understand customer behavior and prepare more accurate inventory plans. We can compare actual vs forecasted sales and use the data for operational planning.

Forecast table with future predicted TotalSales values for the next periods.